SQLite database /Security

*SQLite Database will have **2 way authentication***

1. SQLite database will be accessed by the password: identifierbyvendor+static salt.

2. all the rows/data will be encrypted by the decrypted master key.
3. Encryption will be done by ABP encryption

Static Salt: will be an String identifier
1.All config key values in the config table need to be encrypted with a static salt.
2. The calculator code can be stored with the standard config store, so only encrypt with the static salt.

Prevent application to jel break and reverse engeeniering

1. Jail brea detection
2. take care about reverse engeneering
3. Have to disable NScache

Security with Master key

Generating Masterkey:
When the user will come first time on the app we'll generate a master key that will encrypt with the user password

Change password:
If the user will change the password then we need to decrypt master key with current password and encrypt it with the new password

Use of Master key:
XMPP Credentials need to be encrypted with the static salt and then with the decrypted master_key.

Use of Master key:
Message, Media and the other data *in the DB will be encrypted by the decrypted master key*

Use of Master key:
OMEMO private keys must also be encrypted with the decrypted master key.

Login/Chat security

After Activation user will have only limited number of attempts for wrong password

if user will reach the number of attempts then user will have to enter a captcha after tha user will have only one chance to enter the correct password.

*User also enter a **panic keyword** by tapping that all the data of the app will be removed.*

Messages/contacts/everything from vault/picture/voicememo – when you sent this it must be encrypted with omemo..

Vault Security

For the vault setup user must have to setup a password after that new master key will be generated and encrypt with the vault password.